User Guide For AgentForce CRM

- By Shahzada Moon

AgentForce CRM

Multi-Agent AI System for Automated CRM Data Completion

Web Application Link : https://agentforce-ai-crm.onrender.com

Created by : Shahzada Moon

Team : MOON-Lab (Metadata Optimization & Orchestration Network Lab)

Participation Type : Solo Participant

🏆 Microsoft AI Unlocked 2026 — Track 4: Agent Teamwork

TRACK 4 AGENT TEAMWORK | AZURE AI FOUNDRY | 404 badge not found

SALESFORCE CONNECTED | PYTHON 3.11+

Overview

AgentForce CRM is a **multi-agent AI system** designed to automatically detect and

complete missing Salesforce CRM fields.

Instead of manual data entry, three intelligent AI agents collaborate to analyze context, retrieve historical data, and update CRM records automatically.

The three cooperating agents

Agent	Role
Planner	Detect missing fields and create strategy
Retriever	Query Salesforce history and find similar deals
Executor	Update Salesforce records and log results

Average system confidence: **88%**

⚠️ The Problem

CRM data quality is a major issue across organizations.

Problem	Impact
91% CRM data incomplete	Poor reporting and forecasting
Manual entry takes 5–7 hours/week	Lost productivity
30% annual data decay	Outdated decision making
Incomplete opportunities	Slower deal cycles

Example feedback from sales teams:

> "I spend more time updating CRM than actually selling."

Solution

AgentForce CRM automatically fills missing CRM fields by combining reasoning, retrieval, and execution.

Workflow:

1⃣ Detect missing fields

2 Retrieve historical opportunity data

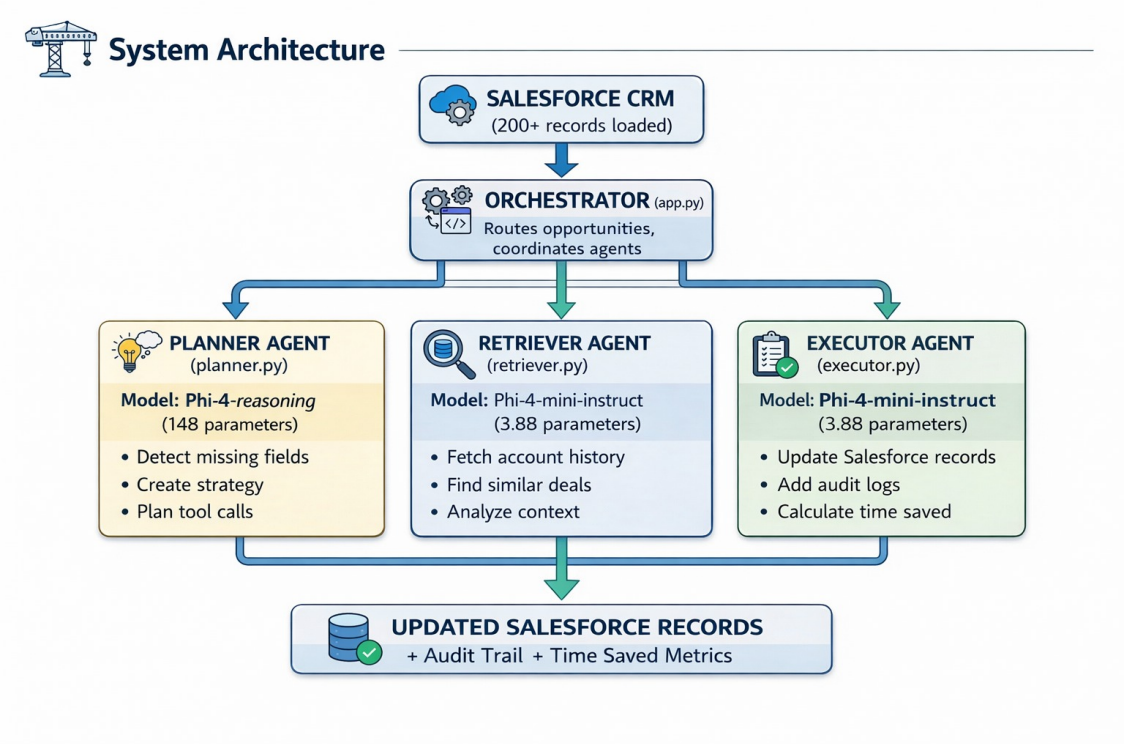
3 Analyze similar deals

4 Infer likely field values

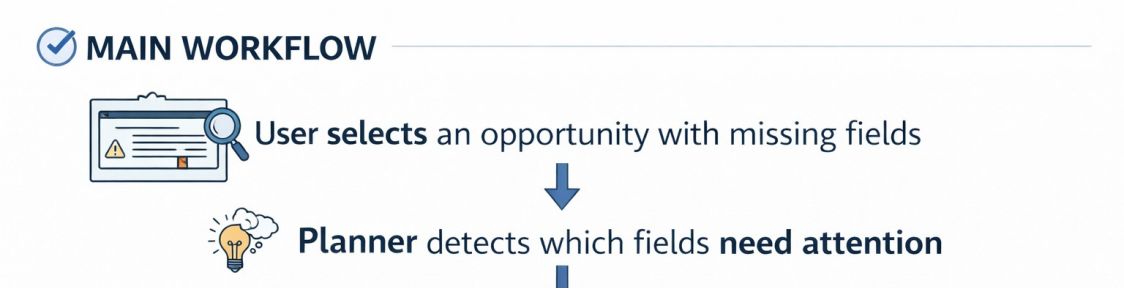
5 Update Salesforce records

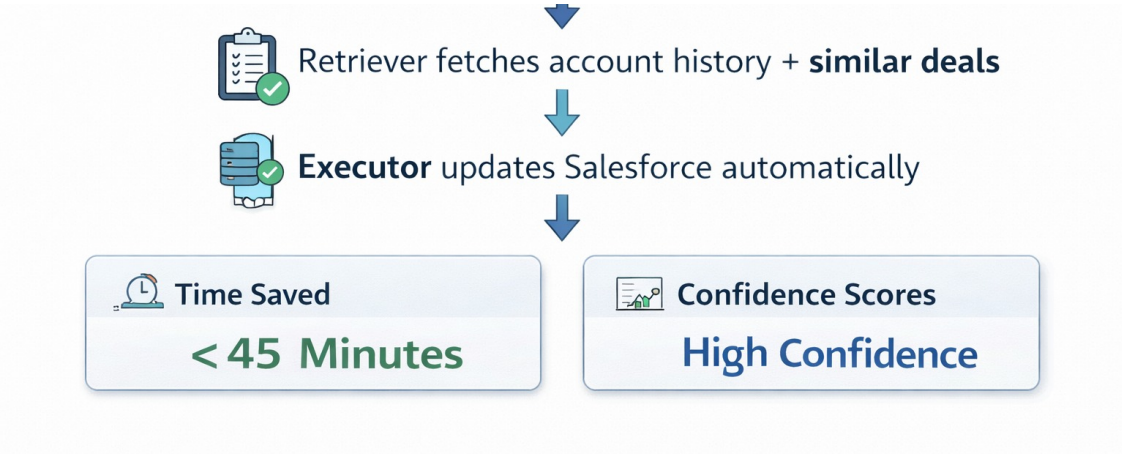
6 Log audit trail and metrics

System Architecture

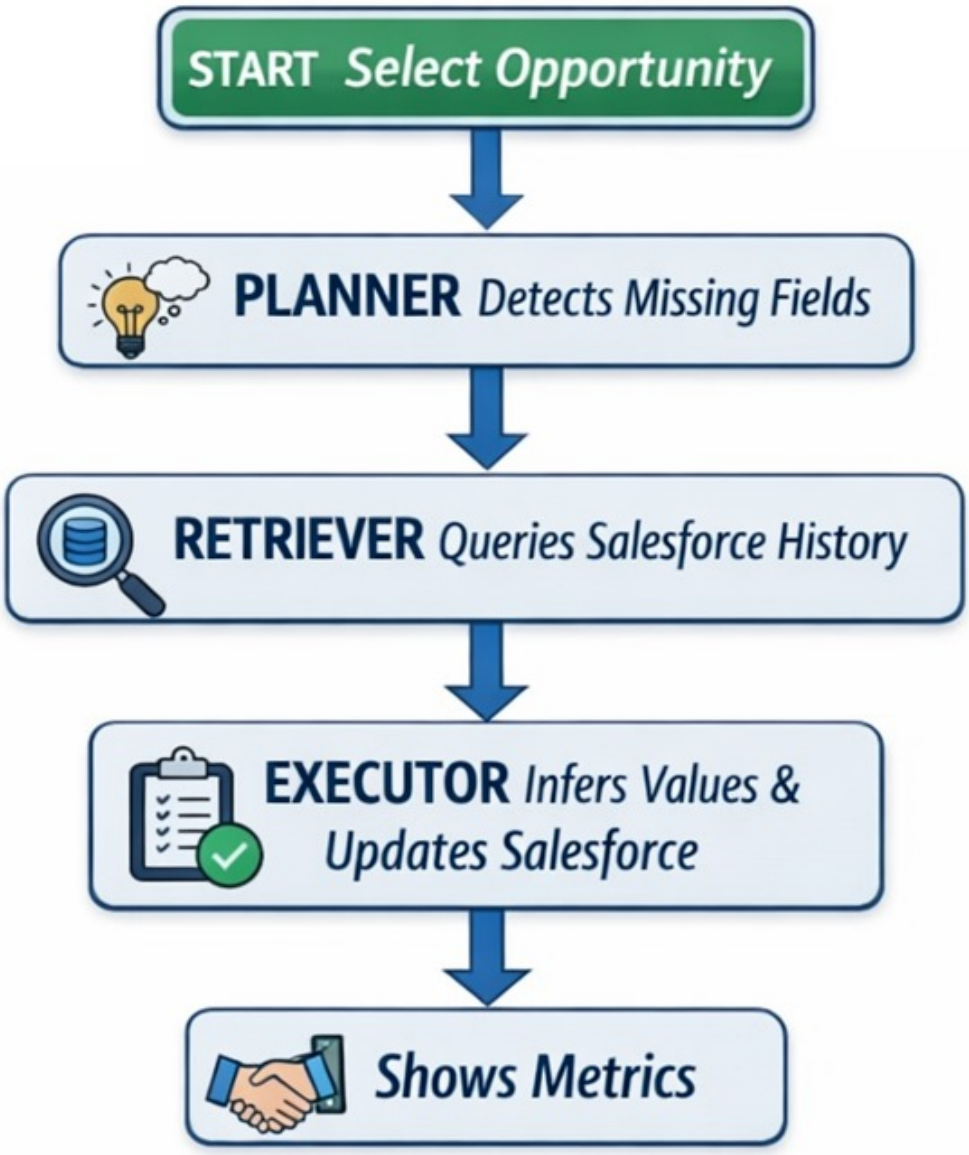


Main WorkFlow



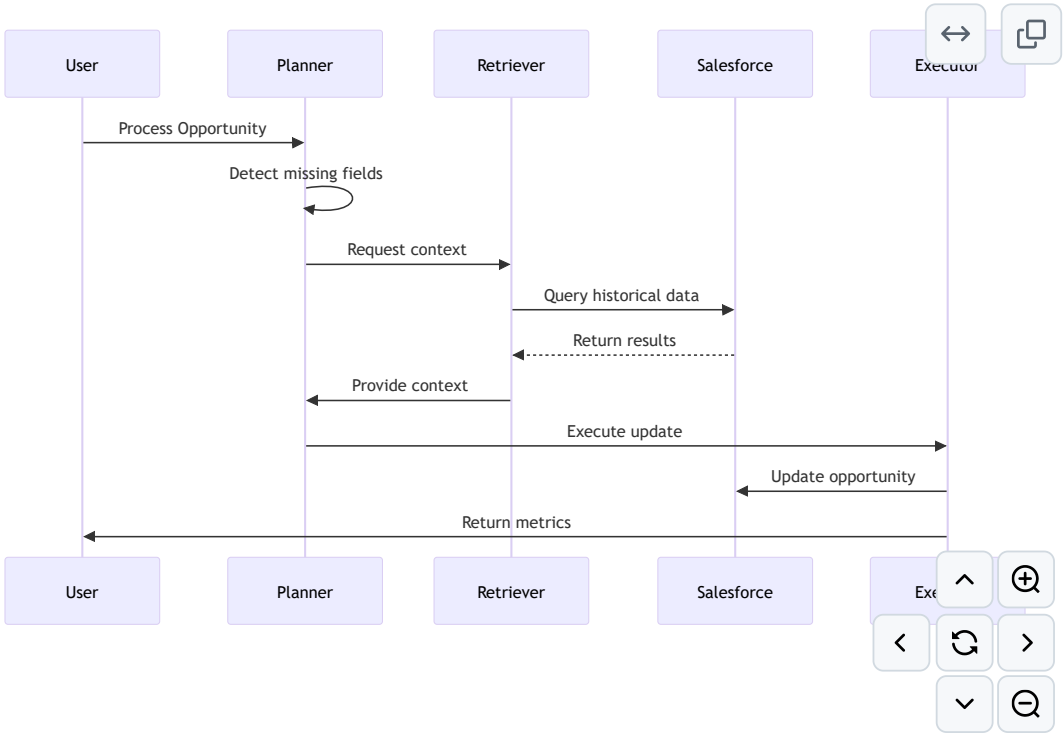


Step-By-Step User WorkFlow



Total Time: < 4 seconds **Value: 45 minutes saved**

Agent Communication



Technology Stack

Layer	Technology
AI Models	Phi-4-reasoning, Phi-4-mini
Platform	Azure AI Foundry
API Layer	Azure OpenAI SDK
Backend	Python

Layer	Technology
CRM Integration	Salesforce API
Deployment	Docker Ready

Microsoft Technologies Used

1 Azure AI Foundry

2 Phi-4-reasoning model

3 Phi-4-mini-instruct model

4 Azure OpenAI SDK

5 Azure AI Content Safety

Salesforce Integration

The system connects to Salesforce using a **Connected App and OAuth2 authentication**.

Example connection code:

```
from simple_salesforce import Salesforce
import os

sf = Salesforce(
    username=os.getenv("SF_USERNAME"),
    password=os.getenv("SF_PASSWORD") + os.getenv("SF_SECURITY_TOKEN"),
    domain="test"
)

accounts = sf.query("SELECT Id FROM Account")
print(f"Connected! {len(accounts['records'])} accounts found")
```

Installation

Clone the repository:

```
git clone https://github.com/23f2002668/agentforce-crm.git
cd agentforce-crm
```

Create virtual environment:

```
python -m venv venv
source venv/bin/activate
```

Install dependencies:

```
pip install -r requirements.txt
```

Environment Configuration

Create `.env`

```
AZURE_FOUNDRY_ENDPOINT=
AZURE_FOUNDRY_API_KEY=

PLANNER_MODEL=Phi-4-reasoning
RETRIEVER_MODEL=Phi-4-mini-instruct
EXECUTOR_MODEL=Phi-4-mini-instruct

SF_USERNAME=
SF_PASSWORD=
SF_SECURITY_TOKEN=
SF_DOMAIN=test
SF_CONSUMER_KEY=
SF_CONSUMER_SECRET=
```

Project Structure

```
AGENTFORCE-CRM/
│
├── 📄 .env                                    # Environment variables
├── 📄 .gitignore                              # Git ignore file
├── 📄 README.md                               # Project documentation
├── 📄 requirements.txt                        # Python dependencies
```

```
├── 📄 Procfile                                    # Render deployment con
├── 📄 render.yaml                                 # Render blueprint conf
│
├── 📄 app.py                                      # Main orchestrator wit
├── 📄 demo.py                                     # CLI demo script
├── 📄 config.py                                   # Configuration managem
├── 📄 test_sf.py                                  # Salesforce connection
│
├── 📄 web_app.py                                  # 🌐 PROFESSIONAL FLASK
│
├── 📂 agents/                                      # Agent modules
│   ├── 📄 __init__.py
│   ├── 📄 planner.py                              # Planner Agent logic
│   ├── 📄 retriever.py                            # Retriever Agent logic
│   └── 📄 executor.py                             # Executor Agent logic
│
├── 📂 tools/                                       # Utility modules
│   ├── 📄 __init__.py
│   └── 📄 salesforce_client.py                    # Salesforce API client
│
├── 📂 data/                                        # Sample data (optional
│    └── 📄 sample_data.csv                        # Sample accounts for S
│
└── 📂 images/                                      # Sample data (optional
    ├── 📄 demo-1.png                              # Output in terminal pa
    ├── 📄 demo-2.png                              # Output in terminal pa
    ├── 📄 web-output.png                          # Output of web applica
    ├── 📄 main-workflow.png                       # Workflow of applicati
    ├── 📄 step-by-step-user-workflow.png          # User - Workflow of ap
    └── 📄 system-architecture.png                 # System architecture o
```

Demo

Run the demo locally:

```
python demo.py
```

Example output:

```
Processing Opportunity: OPP-001

Planner: Missing fields detected
Retriever: Retrieved historical opportunities
Executor: Updated Salesforce record
```

```
Fields completed: 3
Time saved: 45 minutes
Confidence: 88%
```

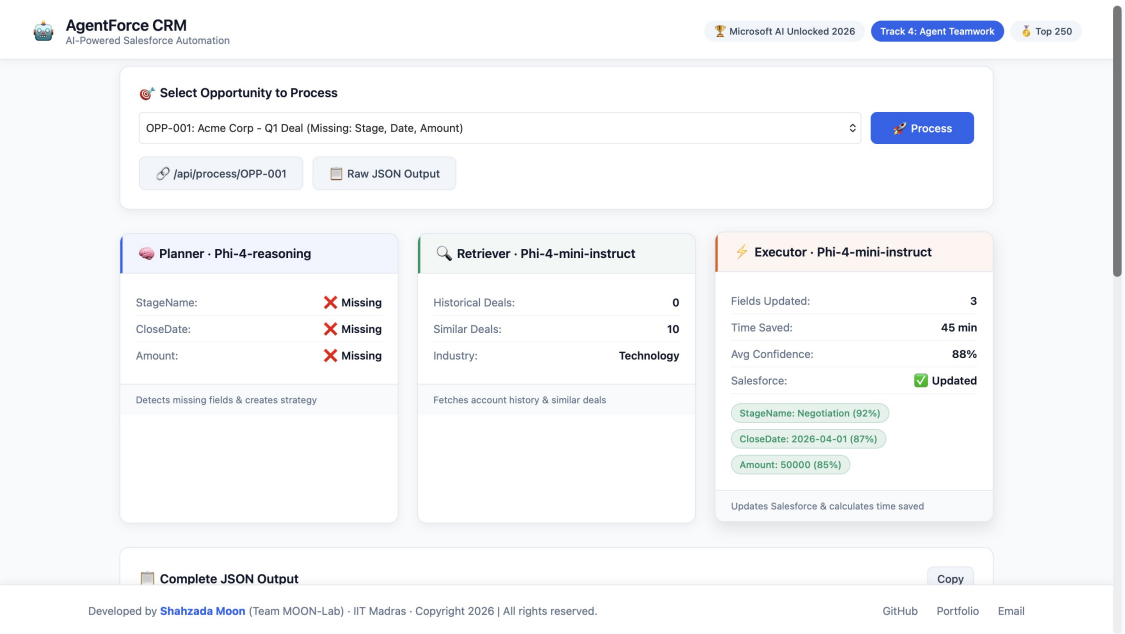
Terminal Output :-

```
[(myenv) bash-5.3$ python3 demo.py

============================================================
🎯 AGENTFORCE CRM - DEMO PRESENTATION
============================================================
============================================================
🚀 AgentForce CRM - Three-Agent System
============================================================

✅ Azure Configuration loaded:
  Endpoint: https://agentforce-crm-resource.services.ai.azure.com/api/projects/AgentForce-CRM
  Planner: Phi-4-reasoning
  Retriever: Phi-4-mini-instruct
  Executor: Phi-4-mini-instruct
✅ Connected to Salesforce
  Initializing Planner Agent...
  ✅ Using REAL Azure OpenAI
  Initializing Retriever Agent...
  Initializing Executor Agent...
✅ All agents initialized
============================================================

📌 DEMO: Processing OPP-001 (Acme Corp - Q1 Deal)
This opportunity has missing: stage, close_date, amount
----------------------------------------

📋 PROCESSING OPPORTUNITY: OPP-001
----------------------------------------

🤔 PLANNER: Analyzing opportunity OPP-001...
  ✅ Planner detected 3 missing fields: StageName, CloseDate, Amount

🔍 RETRIEVER: Fetching context...
  ✅ Retriever found: 0 historical deals, 10 similar deals

⚡ EXECUTOR: Executing plan...
  ✅ Executor updated 3 fields, saved 45 minutes, 88% avg confidence

⏱️Time taken: 3.31 seconds

========================================
📊 PERFORMANCE METRICS
========================================
Fields auto-completed: 3
Time saved: 45 minutes
Average confidence: 88%
Salesforce updated: ✅ YES

========================================
🔄 AGENT COLLABORATION
========================================
1. Planner Agent: Detected missing fields and created strategy
2. Retriever Agent: Fetched account history and similar deals
3. Executor Agent: Updated records and calculated savings

✅ DEMO COMPLETE - All three agents worked together!
============================================================
```

```
📋 Full Result:
{
  "opportunity_id": "OPP-001",
  "planner": {
    "missing_fields": [
      "StageName",
      "CloseDate",
      "Amount"
    ]
  },
  "retriever": {
    "account_history_count": 0,
    "similar_deals_count": 10,
    "industry": "Technology"
  },
  "executor": {
    "updates_made": [
      {
        "field": "StageName",
        "value": "Negotiation",
        "confidence": 92,
        "reason": "Based on similar deals in industry"
      },
      {
        "field": "CloseDate",
        "value": "2026-04-01",
```

```
      "value": "2026-04-01",
      "confidence": 87,
      "reason": "87% confidence based on sales cycle"
    },
    {
      "field": "Amount",
      "value": 50000,
      "confidence": 85,
      "reason": "Based on account history and similar deals"
    }
  ],
  "fields_updated": 3,
  "time_saved": {
    "minutes_saved": 45,
    "hours_saved": 0.75,
    "message": "Saved 45 minutes (3 fields \u00d7 15 min each)"
  },
  "avg_confidence": 88,
  "salesforce_updated": true
 }
}
```


Web Application Output :-



JSON Output From API End-Point


```
{
  "executor": {
    "avg_confidence": 88,
    "fields_updated": 3,
    "salesforce_updated": true,
    "time_saved": {
      "hours_saved": 0.75,
      "message": "Saved 45 minutes (3 fields × 15 min each)",
      "minutes_saved": 45
    },
    "updates_made": [
      {
        "confidence": 92,
```

```
        "field": "StageName",
        "reason": "Based on similar deals in industry",
        "value": "Negotiation"
      },
      {
        "confidence": 87,
        "field": "CloseDate",
        "reason": "87% confidence based on sales cycle",
        "value": "2026-04-01"
      },
      {
        "confidence": 85,
        "field": "Amount",
        "reason": "Based on account history and similar deals",
        "value": 50000
      }
    ]
  },
  "opportunity_id": "OPP-001",
  "planner": {
    "missing_fields": [
      "StageName",
      "CloseDate",
      "Amount"
    ]
  },
  "retriever": {
    "account_history_count": 0,
    "industry": "Technology",
    "similar_deals_count": 10
  }
}
```


Performance Metrics

Metric	Value
Fields completed	3 per opportunity
Time saved	45 minutes
Confidence	88%
Similar deals analyzed	10+
Workflow steps	15

Business Impact

Example for **100 sales representatives**

Metric	Value
Daily time saved	75 hours
Annual hours saved	18,750
Estimated value	₹1.87 Crore per year

Author

Shahzada Moon

Bachelor of Science (B.S.) Data Science And Applications from Indian Institute of Technology Madras

B.Tech (Computer Science & Engineering) from Dr. A. P. J. Abdul Kalam Technical University

GitHub : https://github.com/23f2002668

Portfolio : https://23f2002668.github.io/Portfolio/

Email : 23f2002668@ds.study.iitm.ac.in

Location : Ghaziabad, Uttar Pradesh, India

Final Status

Requirement	Status
Three Agents	✅
Salesforce Integration	✅
Demo Working	✅
Performance Metrics	✅

Requirement	Status
README Complete	✅

Built using Microsoft Azure AI + Salesforce APIs

Thank you for reading my user guide !